

GENERIC LOTKA–VOLTERRA TREE SYSTEMS DETERMINE THEIR TREES

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ABSTRACT. Van der Kamp conjectured that generic homogeneous Lotka–Volterra tree systems associated with nonisomorphic trees are not equivalent under linear changes of variables, and reported a verification for tree orders below nine. We prove the conjecture for every order over a characteristic-zero field. The key point is that the complete projective configuration of generic linear Darboux polynomials consists exactly of the coordinate forms and the edge-supported forms. After diagonal rescaling this is the reduced incidence configuration of the cone over the tree. Its three-element circuits form a 3-uniform hypergraph that reconstructs the underlying tree. We also provide an exact finite certificate for all tree orders from two through nine. A standard-library verifier independently enumerates all Prüfer words, recomputes two projective invariants, and confirms all 47 order-nine cases. Special parameter subvarieties on which extra Darboux polynomials appear are outside the generic-class statement and are described explicitly.

1. INTRODUCTION

A homogeneous n -component Lotka–Volterra (LV) system has the form

$$(1) \quad \dot{x}_i = x_i \sum_{j=1}^n A_{ij} x_j, \quad i = 1, \dots, n.$$

Earlier work associated to every tree T on n vertices a $(3n - 2)$ -parameter class of systems (1) with $n - 1$ additional linear Darboux polynomials, one for every edge [1, 3]. Van der Kamp subsequently studied linear changes of Darboux coordinates and the induced equivalence relation on admissible hypergraphs [2]. Conjecture 12 of that paper states that nonisomorphic trees are not LV-equivalent; the paper reports a complete verification for $n < 9$.

The qualification “generic” is essential. The version of record treats general parametric classes and explicitly does not classify special subclasses [2]. We make this convention exact by working over the rational function field of the $3n - 2$ independent parameters. Our main result is stronger than the first unverified finite order.

Theorem 1.1. *Let T_1, T_2 be finite simple trees of the same order $n \geq 2$. Over a characteristic-zero field, if their generic homogeneous LV tree classes*

Date: August 29, 2026.

are linearly LV-equivalent, allowing a birational parameter remapping, then $T_1 \cong T_2$.

The proof uses neither the integrals of the system nor spectral data. It classifies all generic linear Darboux polynomials, regards their coefficient vectors projectively, and uses only their three-element linear dependences. Section 5 gives a second, finite exact route for orders up to nine and independently reproduces the previously reported range.

2. GENERIC TREE CLASSES AND EQUIVALENCE

Let $T = (V, E)$ be a tree with $V = \{1, \dots, n\}$. For each vertex v and each oriented incidence $u \rightarrow v$ of an edge uv , introduce algebraically independent parameters

$$a_v, \quad d_{u,v}.$$

There are $n + 2(n - 1) = 3n - 2$ of them. We work over

$$\mathbb{K}_T = \mathbb{Q}(a_v \ (v \in V), d_{u,v} \ (uv \in E)).$$

The interaction matrix A_T is specified by $A_{vv} = a_v$, $A_{uv} = d_{u,v}$ when $uv \in E$, and

$$(2) \quad A_{ij} = A_{kj} \quad (ik \in E, \ j \notin \{i, k\}).$$

This is the standard generic tree-system parametrization [3]. Write

$$F_T(x) = \text{diag}(x)A_Tx.$$

A nonzero polynomial P is a Darboux polynomial for F_T if $\dot{P} = CP$ for a polynomial cofactor C . We identify nonzero linear Darboux polynomials that differ by a nonzero scalar and write $[P]$ for the corresponding projective class.

Definition 2.1. Two generic tree classes are linearly LV-equivalent if, after a birational identification of their parameter fields, there is an $M \in \text{GL}_n$ over the common field such that

$$(3) \quad F_{T_2}(y) = MF_{T_1}(M^{-1}y).$$

Specializations that impose a nonzero algebraic relation among the generic parameters do not witness this class equivalence.

Only the necessary consequence of Definition 2.1 is used: pullback by M bijects the projective linear Darboux polynomials of the two generic vector fields and preserves all linear dependences.

3. CLASSIFICATION OF GENERIC LINEAR DARBOUX POLYNOMIALS

The row equalities have a useful path description.

Lemma 3.1. *If $i \neq j$ and r is the neighbor of j on the unique path from j to i , then*

$$A_{ij} = d_{r,j}.$$

Proof. In the component of $T - j$ containing i , relation (2) makes column j constant along every edge. Its value therefore equals $A_{rj} = d_{r,j}$. $\square$

We use the linear-Darboux criterion proved in [2, Lemma 2]. If

$$(4) \quad P = \sum_{i \in I} \alpha_i x_i, \quad \alpha_i \neq 0,$$

has exact support I , then necessarily

$$(5) \quad A_{ij} = A_{kj} \quad (i, k \in I, j \notin I),$$

$$(6) \quad \alpha_i(A_{ij} - A_{jj}) = \alpha_j(A_{ii} - A_{ji}) \quad (i, j \in I).$$

For homogeneous systems these conditions, together with the displayed nonzero coefficients, are also sufficient.

Lemma 3.2. *At the generic point, all projective linear Darboux polynomials are*

$$(7) \quad \mathcal{D}_T = \{[x_v] : v \in V\} \cup \{[P_{uv}] : uv \in E\},$$

where, up to nonzero scalar,

$$(8) \quad P_{uv} = (d_{v,u} - a_u)x_u + (a_v - d_{u,v})x_v.$$

In particular, $|\mathcal{D}_T| = 2n - 1$.

Proof. The singleton supports give the coordinate Darboux polynomials. Suppose that $|I| \geq 2$ in (4). If $T[I]$ is disconnected, choose two closest components and vertices $p, q \in I$ whose connecting path has an interior vertex $j \notin I$. The first neighbors r, s of j toward p, q are distinct. Lemma 3.1 gives

$$A_{pj} = d_{r,j}, \quad A_{qj} = d_{s,j}.$$

The right sides are distinct algebraically independent parameters, contradicting (5). Hence $T[I]$ is connected.

If $|I| = 2$, the support is an edge uv . Equation (6) determines a unique projective coefficient ratio and gives (8).

If $|I| \geq 3$, the tree $T[I]$ contains two edges pq, qr with $p \neq r$. Put

$$\begin{aligned} X &= d_{p,q} - a_q, & Y &= a_p - d_{q,p}, \\ Z &= d_{q,r} - a_r, & W &= a_q - d_{r,q}. \end{aligned}$$

Lemma 3.1 and (6), for the pairs $(p, q), (q, r), (p, r)$, give

$$\alpha_p X = \alpha_q Y, \quad \alpha_q Z = \alpha_r W, \quad \alpha_p Z = \alpha_r Y.$$

Multiplying the first equality by Z and substituting the other two yields $\alpha_r Y(X - W) = 0$. Both α_r and Y are nonzero in the rational function field, so

$$d_{p,q} + d_{r,q} - 2a_q = 0.$$

This contradicts algebraic independence in characteristic zero. Thus no support of size at least three occurs generically. $\square$

Remark 3.3 (The special boundary). The last obstruction is real, not an artifact of cancellation. On a three-vertex path $p - q - r$, the coefficient matrix of the three equations in the preceding proof has determinant

$$-(a_p - d_{q,p})(a_r - d_{q,r})(2a_q - d_{p,q} - d_{r,q}).$$

Consequently a higher-support Darboux polynomial can appear on a proper algebraic parameter hypersurface. This explains why Theorem 1.1 is stated for general/generic classes and not for arbitrary specializations.

4. THE CONE CONFIGURATION AND RECONSTRUCTION

Write $P_{uv} = \lambda_{uv,u}x_u + \lambda_{uv,v}x_v$; both coefficients are nonzero in $\mathbb{K}_T$. Root T at v_0 , set $s_{v_0} = 1$, and, for a parent–child edge uv , define recursively

$$(9) \quad s_v = -\frac{\lambda_{uv,v}}{\lambda_{uv,u}}s_u.$$

Because T has no cycle, these scalings have no compatibility obstruction. In coordinates $z_v = s_v x_v$, after independently rescaling projective forms,

$$(10) \quad \mathcal{D}_T \cong \{[z_v] : v \in V\} \cup \{[z_u - z_v] : uv \in E\}.$$

Let G_T be the cone over T : add a vertex 0 and all spokes $0v$. The vectors in (10) are the columns of an oriented incidence matrix of G_T after deleting the row indexed by 0. A set of these columns is minimally dependent exactly when the corresponding graph edges form a simple cycle. Indeed, an oriented cycle has signed column sum zero, whereas the columns of a forest are independent by successive leaf elimination.

The three-element circuits are therefore exactly

$$(11) \quad H_{uv} = \{[x_u], [P_{uv}], [x_v]\}, \quad uv \in E,$$

because every triangle in G_T consists of a tree edge and its two spokes. Let $\mathcal{H}_T$ be the 3-uniform hypergraph with ground set $\mathcal{D}_T$ and hyperedges (11).

Lemma 4.1. *The unlabeled hypergraph $\mathcal{H}_T$ determines the unlabeled tree T .*

Proof. For every vertex and edge,

$$\deg_{\mathcal{H}_T}([x_v]) = \deg_T(v), \quad \deg_{\mathcal{H}_T}([P_e]) = 1.$$

Assume $n \geq 3$, and let J be the hypergraph elements of degree at least two. They are precisely the coordinate forms associated with nonleaf vertices. No edge of a tree of order at least three joins two leaves. Hence each hyperedge meets J in two elements, for an internal–internal edge, or one element, for an internal–leaf edge. Join the two elements in the first case; in the second, attach one fresh leaf to the unique element. The resulting graph is T up to relabeling. For $n = 2$, the only tree is K_2 . $\square$

Proof of Theorem 1.1. An invertible linear conjugacy pulls a linear Darboux polynomial of the target back to one of the source; its inverse gives the reverse map. By Lemma 3.2, it induces a projective bijection $\mathcal{D}_{T_1} \rightarrow \mathcal{D}_{T_2}$.

Invertible linear maps preserve and reflect linear dependence, so the bijection restricts to an isomorphism $\mathcal{H}_{T_1} \cong \mathcal{H}_{T_2}$. Lemma 4.1 yields $T_1 \cong T_2$. $\square$

5. EXACT FINITE CERTIFICATION THROUGH ORDER NINE

Although Theorem 1.1 is noncomputational, we independently checked the requested first unreported order and reproduced the earlier baseline. A serialized certificate contains every unlabeled tree of orders two through nine as an integer edge list. It records two exact invariants.

The first is the reconstruction in Lemma 4.1. For the second, note that every circuit of G_T is indexed by an unordered pair $u, v \in V$: it consists of the two spokes $0u, 0v$ and the unique $u-v$ path in T . For each of the $2n - 1$ configuration elements, count the circuits of every cardinality that contain it, then take the multiset of these profiles. Projective equivalence preserves this circuit-incidence invariant.

Proposition 5.1 (Exact computational proposition). *For each $2 \leq n \leq 9$, the circuit-incidence invariant just defined is pairwise distinct on all unlabeled trees of order n . The exact counts and numbers of distinct signatures are as follows.*

n	2	3	4	5	6	7	8	9
<i>unlabeled trees</i>	1	1	2	3	6	11	23	47
<i>distinct signatures</i>	1	1	2	3	6	11	23	47

The discovery program uses NetworkX only to generate representatives. The independent verifier imports neither NetworkX nor the discovery module. It enumerates every word in $\{0, \dots, n - 1\}^{n-2}$, decodes each Prüfer word, computes an AHU-style free-tree parenthesis code, and checks that the set of codes equals the serialized list. It then recomputes both invariants from the edge lists. All arithmetic and comparisons are integral or symbolic; no floating-point calculation is used. Six deliberately corrupted inputs are rejected, including a missing order-nine tree and forged invariant data.

The full verification command is

```
python code/verifier/verify_tree_certificate.py \
certificates/trees_n2_n9.json
```

The verified certificate SHA-256 is

```
5be053e1f8e676769ad33237055bed21c76f9825fd373f7832fb2d21dc414fe6
```

6. LIMITATIONS AND DISCLOSURE

The theorem concerns generic/general parameter classes. It does not say that every special parameter value has exactly $2n - 1$ projective linear Darboux polynomials; Remark 3.3 gives an explicit algebraic boundary where the generic classification changes. The proof concerns linear conjugacies of homogeneous LV systems and does not classify nonlinear, birational, or time-dependent transformations.

Computer assistance was used for literature search, symbolic checking of the three-vertex obstruction, exact finite enumeration, independent certificate verification, and manuscript preparation. An AI coding and reasoning system assisted with proof exploration, code, and exposition. Every decisive finite claim is recomputed by the disclosed fail-closed verifier, while the general theorem has the human-readable proof above. No Lean, Coq, Isabelle, or other proof assistant was used.

DATA AND CODE AVAILABILITY

The certificate, discovery generator, independent verifier, destructive-input tests, literature ledger, proof audit, and release manifest accompany this article. The verifier is deterministic and uses no random seed.

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